Does Reward Curation Kill Diversity? Systematic Evidence of Surprising Resilience in Iterative Self-Training

Anonymous ACL submission

Abstract

Iterative self-training with reward curation, where a reward model selects the best candidate response at each generation for supervised fine-tuning, is widely expected to erode output diversity through compounding selection pressure. We test this expectation with a controlled factorial experiment spanning 8 configurations, 3 iterative generations, and both Low-Rank Adaptation (LoRA) and full fine-tuning. The central finding contradicts the prevailing concern: semantic diversity, measured by Vendi Score over sentence embeddings, remains within $\pm 1\%$ of its initial value across all curated conditions, even at a 6.25% retention rate (top-1-of-16) where Vendi Score changes by only -0.19%. However, this stability masks what we term *metric-dependent collapse*. Iterative self-training causes substantial format homogenization regardless of curation strategy, with opening unique ratio falling from 0.757 to 0.477 and near-duplicate start rates rising from 37.9% to 75.6% under matched $K{=}4$ conditions, while embedding-based metrics remain blind to this convergence. We further document a direction reversal: curation suppresses diversity under LoRA (-3.40 percentage points relative to a random baseline) but promotes it under full fine-tuning ($+0.63$ pp), a pattern that tracks parameter capacity rather than selection intensity. These results establish that reward curation does not cause semantic collapse in iterative self-training, that diversity audits relying solely on embedding-based measures miss significant structural homogenization, and that conclusions about curation effects depend critically on the training regime under which they are measured.

1 Introduction

Iterative self-training with reward-based curation has become a prominent strategy for language model alignment. The current model generates multiple candidate responses per prompt, a reward model scores them, and the top-scoring response is selected for supervised fine-tuning to produce the next-generation model. This loop raises a fundamental stability question: if each generation retains only a narrow fraction of outputs (1 out of 4 or 1 out of 8 candidates), the compounding effect should progressively narrow the output distribution and erode the breadth that makes language models practical across diverse user needs.

Two research threads support this concern. The model collapse literature demonstrates that training on synthetic outputs leads to progressive distribution narrowing (Shumailov et al., 2024; Alemohammad et al., 2024): rare modes are underrepresented in synthetic samples, this compounds across iterations, and the model collapses onto high-density modes. Separately, RLHF studies document that single-round reward optimization reduces lexical and semantic variety (Kirk et al., 2024). Together, these establish a strong prior: iterative reward-curated self-training should cause diversity collapse with increasing severity at each generation.

Yet neither literature directly addresses the deployed setting. Model collapse results study unfiltered self-consumption where the model trains on its full output distribution; reward-curated pipelines apply an external selection filter, retain a fixed prompt scaffold across generations, and decode stochastically, all of which violate the theoretical preconditions for progressive narrowing. The RLHF literature examines single rounds rather than multi-generation iteration. No prior work has systematically measured whether diversity degrades across multiple generations of reward-curated fine-tuning under controlled factorial conditions.

We fill this gap with a controlled experiment spanning 8 configurations and 3 generations. The central result overturns expectations: semantic diversity (Vendi Score (Friedman and Dieng, 2023)) remains stable across all curated conditions

(-0.27% to $+0.91\%$), including extreme selection pressure (K=16, 6.25% retention, $\Delta = -0.19\%$). However, this stability masks *metric-dependent collapse*: format diversity degrades substantially regardless of curation, with near-duplicate start rate rising from 37.9% to 75.6% and opening entropy dropping 0.75 bits at K=4. Standard embedding-based metrics are blind to this convergence. A second finding concerns the interaction between adaptation method and curation: under LoRA, curation suppresses Vendi Score by -3.40 pp relative to random, but under Full-FT with higher learning rate, curation promotes diversity by $+0.63$ pp. This pattern is consistent with a capacity artifact: LoRA's rank-16 bottleneck limits how much distributional structure the model can absorb, and the apparent suppression disappears when this constraint is removed.

Our experimental design isolates four factors: adaptation method (LoRA versus Full-FT), selection pressure ($K \in \{4, 8, 16\}$, corresponding to 25%, 12.5%, and 6.25% retention rates), learning rate (5×10^{-6} versus 2×10^{-5}), and curation strategy (reward top-1 versus random selection). All experiments use Qwen3-0.6B (596M parameters) as the base model, ArmoRM-Llama3-8B-v0.1 (Wang et al., 2024) as the reward model, and a fixed set of 2,200 prompts from UltraFeedback. This factorial structure enables direct attribution of observed diversity changes to specific experimental factors rather than confounding variation in data composition or model scale.

Our contributions are as follows:

1. We provide systematic evidence that reward curation does not cause semantic diversity collapse in iterative self-training. Across 8 configurations and 3 generations, all curated conditions maintain Vendi Score within 1% of the initial value, even under extreme selection pressure (K=16, 6.25% retention). This result contradicts expectations from the model collapse literature.

2. We identify metric-dependent collapse: format diversity (response openings, structural templates) degrades substantially while embedding-based semantic diversity remains stable. This degradation occurs at comparable rates under curated and random selection, which indicates it is inherent to iterative self-training rather than caused by reward filtering. The effect intensifies with candidate pool size (K=16, -0.19% Vendi but 1.13 bits opening entropy loss at K=8). We propose two complementary indicators, `opening_entropy` and `near_dup_start_rate`, that capture this previously invisible dimension of homogenization, and recommend their inclusion in diversity audits alongside standard embedding-based measures.

3. We observe a direction reversal in the effect of curation across adaptation methods: curation suppresses diversity under LoRA (-3.40 pp) but promotes it under Full-FT ($+0.63$ pp), a pattern consistent with LoRA's rank-16 bottleneck constraining how much distributional structure the model can absorb from curated data.

2 Related Work

Model Collapse and Self-Consuming Training. Training generative models on their own outputs across multiple generations risks progressive quality degradation. (Shumailov et al., 2024) formalized this phenomenon as *model collapse* and showed that iterative retraining on unfiltered synthetic data causes tail distributions to vanish and outputs to converge toward a narrow mode. (Alemohammad et al., 2024) extended this analysis to self-consuming generative models more broadly and established theoretical conditions under which the output distribution contracts with each generation. Building on these findings, DIVE (Qin et al., 2025) proposed diversity-preserving iterative self-improvement and introduced explicit diversity constraints during self-training to counteract distributional narrowing. A common thread in these works is that collapse arises from *unfiltered* self-consumption, where all generated data is fed back without selection. Our setting differs fundamentally: we study *reward-filtered* iterative training, where only top-scoring outputs are retained at each generation. The collapse dynamics under selective pressure are distinct from those under uniform resampling, and whether reward curation compounds into semantic collapse across generations had not been empirically tested prior to our work.

Diversity in RLHF and Reward Optimization. Reward optimization is known to reduce output variability in single-generation settings. (Kirk et al., 2024) measured lexical and semantic diversity before and after RLHF and found significant reduc-

tions across multiple diversity metrics within a single alignment round. CD-RLHF (Sun et al., 2025) proposed curiosity-driven exploration bonuses to counteract this effect; these bonuses encourage the policy to maintain diverse behavior during reward maximization. These studies share a limitation: they observe diversity changes within a single training round and do not examine whether the effect compounds when the curated outputs become training data for subsequent generations. Our experiments span three iterative generations of reward-filtered self-training and directly test the compounding hypothesis. We find that semantic diversity (measured by embedding distances and Vendi Score) remains stable across generations. This result contradicts the extrapolation from single-round observations to iterative collapse.

Diversity-Preserving Methods. Several recent methods explicitly optimize for output diversity alongside reward. DARLING (Li et al., 2025a) introduces diversity-aware reward learning that penalizes reward models for assigning high scores to repetitive outputs. GEM (Li et al., 2025b) frames diversity preservation as a game-theoretic problem and trains a population of policies that are jointly rewarded for collective coverage of the output space. These methods share a premise: that standard reward optimization inevitably produces homogeneous outputs and that active intervention is required to maintain diversity. Our results challenge this premise at the semantic level. Under standard reward curation without any diversity-preserving mechanism, we observe no measurable semantic collapse across three generations. The homogenization we do observe operates at the format and template level (opening patterns, structural repetition), a dimension these methods do not explicitly target. This suggests that diversity-preserving methods may be optimizing against a failure mode that does not manifest in practice at the semantic layer, while the actual homogenization occurs in a dimension they leave unaddressed.

Diversity Measurement. Standard diversity metrics in text generation fall into two categories: reference-free surface metrics such as Self-BLEU (Zhu et al., 2018), which measures n-gram overlap across generated samples, and kernel-based measures such as the Vendi Score (Friedman and Dieng, 2023), which computes an effective number of distinct modes from a similarity kernel over embeddings. Both approaches capture semantic or

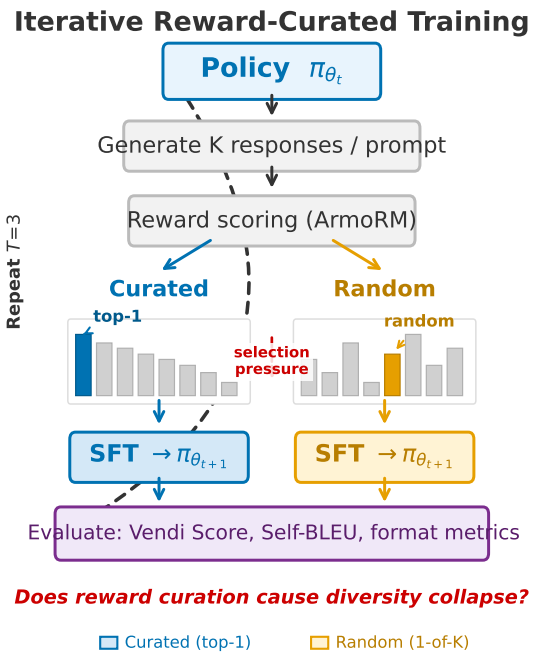


Figure 1: Iterative self-training pipeline. Each generation generates K candidates per prompt, scores them with ArmoRM, selects the top-1, and fine-tunes the base model for the next generation.

lexical variety but are blind to structural homogenization: two responses can have entirely different content (low Self-BLEU, high embedding distance) yet open with identical templates and follow the same organizational pattern. Our analysis reveals this blind spot empirically. We introduce two complementary surface metrics, opening entropy and near-duplicate start rate, that specifically measure template-level repetition invisible to embedding-based evaluation. These metrics expose the format collapse that standard diversity measures miss, and they explain why prior work could simultaneously report stable diversity scores and yet produce outputs that feel increasingly formulaic to human evaluators.

3 Experimental Setup

3.1 Iterative Self-Training Pipeline

We study diversity dynamics through an iterative self-training loop (Figure 1). The base model is Qwen3-0.6B (Qwen Team, 2025) (596M parameters), chosen as the smallest model capable of coherent instruction-following in tractable compute budgets. We use UltraFeedback (Cui et al., 2024) as the prompt source, sampling 2,200 gen-

Table 1: Experimental configurations. C1–C4 each has a matched random baseline (C5–C7); C8 extends the selection pressure range without a paired baseline. Trainable % refers to the fraction of model parameters updated during fine-tuning.

ID	Adaptation	K	LR	Curation	Trainable
C1	LoRA (r=16)	4	2e-4	ArmoRM top-1	1.67%
C2	LoRA (r=16)	8	2e-4	ArmoRM top-1	1.67%
C3	Full-FT	4	5e-6	ArmoRM top-1	100%
C4	Full-FT	4	2e-5	ArmoRM top-1	100%
C8	Full-FT	16	2e-5	ArmoRM top-1	100%
C5	LoRA (r=16)	4	2e-4	Random 1-of-K	1.67%
C6	Full-FT	4	5e-6	Random 1-of-K	100%
C7	Full-FT	4	2e-5	Random 1-of-K	100%

eral instruction-following prompts (2,000 train, 200 eval). Each generation proceeds in four steps: (1) generate K candidate responses per prompt; (2) score with ArmoRM-Llama3-8B-v0.1 (Wang et al., 2024); (3) select the top-1 candidate; (4) fine-tune the base model on the selected subset. We run three complete generations (Gen 0 $\rightarrow$ Gen 1 $\rightarrow$ Gen 2). Gen 0 metrics reflect the model after the first training round, not the unmodified base model. All candidate generation uses temperature T=0.7.

3.2 Experimental Conditions

Table 1 summarizes the eight configurations. Our design isolates four factors through controlled comparisons. First, adaptation method (LoRA vs. Full Fine-Tuning) isolates the effect of parameter capacity: LoRA (Hu et al., 2022) updates only 1.67% of parameters (rank 16), while Full-FT (Full Fine-Tuning) updates all parameters. Second, candidate pool size ($K \in \{4, 8, 16\}$) isolates selection pressure, corresponding to 25%, 12.5%, and 6.25% retention rates respectively. Third, learning rate (5e-6 vs. 2e-5 for Full-FT) isolates learning intensity across a conservative to aggressive optimization range. Fourth, curation strategy (ArmoRM top-1 vs. random selection) isolates the causal effect of reward-based filtering: configurations C5, C6, and C7 serve as matched baselines for C1, C3, and C4 respectively and differ only in whether selection is reward-guided or random.

This factorial structure enables direct attribution. If reward curation causes diversity collapse, curated conditions (C1–C4, C8) should show steeper diversity decline than their random counterparts (C5–C7) across generations. If parameter capacity mediates collapse, LoRA and Full-FT conditions should diverge even under identical curation. If selection pressure drives collapse, C8 (K=16, 6.25% retention) should collapse faster than C2 (K=8) and C1 (K=4).

3.3 Evaluation Metrics

We evaluate diversity at two levels: semantic content and surface format.

Semantic diversity. Our primary metric is the Vendi Score (Friedman and Dieng, 2023), a kernel-based diversity measure computed over sentence embeddings from all-MiniLM-L6-v2 (Wang et al., 2020). The Vendi Score equals the exponential of the Shannon entropy of the eigenvalues of the kernel matrix, with higher values indicating greater diversity. We complement this with Self-BLEU (Zhu et al., 2018), defined as the average pairwise BLEU score among all responses in the evaluation set (lower indicates more diversity), and Nearest-Neighbor Distance (NND), the average cosine distance to the nearest neighbor in embedding space (higher indicates more diversity). All semantic metrics are computed over the 200 evaluation prompts with one response per prompt per generation.

Format diversity. Standard embedding-based metrics can miss surface-level homogenization because semantically distinct responses may share identical templates. We introduce two format-level indicators. Opening Entropy measures the Shannon entropy of first-10-token sequences across responses; lower values indicate convergence toward fixed opening patterns. Near-Duplicate Start Rate measures the fraction of response pairs sharing a 50-character prefix; higher values indicate template repetition. These metrics capture a distinct failure mode: models that produce varied content but wrap it in repetitive formatting.

Quality and training dynamics. We track mean ArmoRM score per generation to verify that quality does not degrade, and per-generation training loss to monitor learning dynamics and detect underfitting or overfitting.

3.4 Training Details

Both LoRA and Full-FT configurations use AdamW (Loshchilov and Hutter, 2019) with cosine scheduling and train for 2 epochs per generation (3 for C4/C8). LoRA uses rank 16, α=32, applied to all linear layers. All experiments run on 2$\times$ NVIDIA RTX 5090 GPUs. Complete hyperparameters are in Appendix Table 5.

Table 2: Vendi Score across three self-training generations. Δ denotes relative change from Gen 0 to Gen 2. All curated conditions remain within $\pm 1\%$, while random baselines show larger positive drift.

	Config	Gen 0	Gen 1	Gen 2	Δ (%)
Curated (ArmoRM top-1)					
C1	LoRA, K=4	107.54	106.73	107.25	−0.27
C2	LoRA, K=8	107.21	107.26	107.25	+0.04
C3	Full-FT, 5e-6	107.10	107.61	108.07	+0.91
C4	Full-FT, 2e-5	107.81	108.03	108.54	+0.68
C8	Full-FT, K=16	107.32	106.73	107.11	−0.19
Random baseline (1-of-K)					
C5	LoRA, K=4	105.42	108.21	108.72	+3.13
C6	Full-FT, 5e-6	107.17	107.87	108.64	+1.37
C7	Full-FT, 2e-5	108.08	107.85	108.13	+0.05

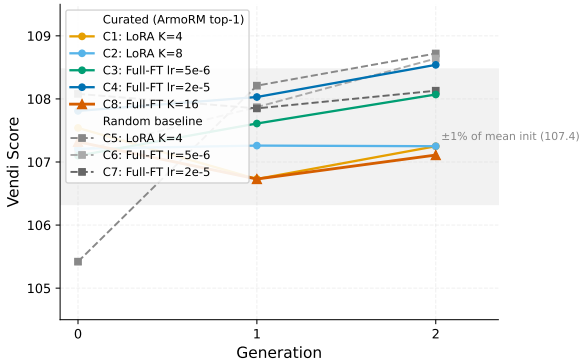


Figure 2: Vendi Score trajectories across three self-training generations for all 8 configurations. Curated conditions (solid lines) remain within $\pm 1\%$ of Gen 0, while random baselines (dashed) show larger positive drift.

4 Results

4.1 No Semantic Diversity Collapse

Reward-based curation does not cause semantic diversity collapse over three generations of iterative self-training. Table 2 reports Vendi Scores for all conditions across generations. Among the five curated configurations (C1–C4, C8), relative changes from Gen 0 to Gen 2 range from -0.27% to $+0.91\%$, all within $\pm 1\%$. This stability persists across every factor we vary: adaptation method (LoRA vs. Full-FT), candidate pool size ($K \in \{4, 8, 16\}$), and learning rate (5e-6 vs. 2e-5). C8, with the strongest selection pressure (K=16, 6.25% retention), produces a change of only -0.19%. Both Full-FT curated conditions (C3, C4) exhibit monotonically increasing Vendi Scores, while LoRA conditions remain flat or fluctuate transiently before recovery.

Figure 2 visualizes these trajectories. Secondary metrics corroborate this result. Self-BLEU stays below 0.01 for all curated conditions (absolute values 0.002–0.009), and Self-BLEU-4 computed over full training corpora confirms this pattern (all conditions $<$0.016; Appendix E). Nearest-Neighbor Distance remains between 0.70 and 0.71 with no systematic contraction. These results contrast with predictions from the model collapse literature (Shumailov et al., 2024; Alemohammad et al., 2024). The critical difference is that each generation draws fresh stochastic samples (T=0.7) from the current model, and the prompt set provides a constant external anchor.

4.2 Relative Suppression, Not Absolute Collapse

Curated conditions do not collapse, but they grow diversity more slowly than matched random baselines. C5 and C6 show Vendi Score increases of $+3.13\%$ and $+1.37\%$ (Table 2), compared to -0.27% and $+0.91\%$ for their curated counterparts C1 and C3. This distinction matters: studies reporting that reward filtering reduces diversity relative to unfiltered baselines (Kirk et al., 2024) are sometimes read as evidence of degeneration. Our data show that a curated model holding steady has not collapsed, even if an uncurated model drifted above it. Evaluating curation requires comparing each model against its own starting point, not against a diverging uncurated trajectory.

4.3 Direction Reversal Across Training Regimes

The curation gap (curated Δ minus random Δ, in percentage points) reverses sign as training capacity increases. Under LoRA (C1 vs. C5), the gap is -3.40 pp. Under Full-FT at 5e-6 (C3 vs. C6), it contracts to -0.46 pp. Under Full-FT at 2e-5 (C4 vs. C7), it reverses to $+0.63$ pp: curation now promotes diversity (Figure 3).

Training loss trajectories (Table 3) explain this reversal. Under LoRA, curated and random conditions converge to similar final loss (C1: 0.166, C5: 0.152). The rank-16 adapter's low-dimensional bottleneck limits how far any signal can reshape the output distribution. Random data, unconstrained by reward consistency, permits slightly more distributional drift within this bottleneck, producing the observed gap. Full-FT removes this constraint. At lr=2e-5, reward-selected data provides varied high-quality completions across topics that translate into greater semantic diversity when the model

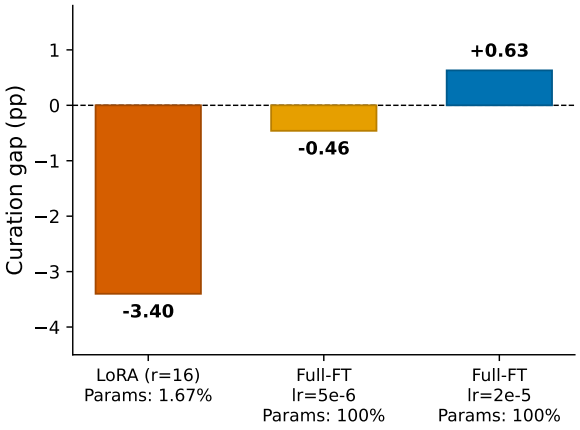


Figure 3: Curation gap (curated Δ minus random Δ, in percentage points) across training regimes. The gap transitions from negative (curation suppresses diversity) under LoRA to positive (curation promotes diversity) under Full-FT with higher learning rate.

Table 3: Training loss per generation. LoRA conditions converge to similar final loss regardless of curation strategy. C8 (K=16) reaches the lowest Gen 2 loss (0.097).

	Config	Gen 0	Gen 1	Gen 2
C1	LoRA, curated	0.413	0.240	0.166
C2	LoRA (K=8), curated	0.411	0.239	0.168
C5	LoRA, random	0.407	0.224	0.152
C3	Full-FT 5e-6, curated	0.461	0.341	0.285
C6	Full-FT 5e-6, random	0.461	0.335	0.277
C4	Full-FT 2e-5, curated	0.380	0.212	0.140
C8	Full-FT 2e-5, K=16	0.206	0.129	**0.097**
C7	Full-FT 2e-5, random	0.377	0.199	0.132

can fully absorb the curated distribution. The progression from LoRA through conservative Full-FT to aggressive Full-FT traces a continuum from parameter-limited to parameter-sufficient training.

This pattern is consistent with a capacity-mediated mechanism: LoRA's rank-16 bottleneck limits how much distributional structure the model can absorb, and the apparent suppression effect disappears when this constraint is removed. The practical implication is that experiments conducted exclusively with parameter-efficient methods may reach conclusions about curation's diversity impact that do not generalize to full fine-tuning.

4.4 Quality-Diversity Non-Tradeoff

Curation produces neither a quality gain large enough to motivate its use nor a diversity cost large enough to discourage it. Mean ArmoRM scores at Gen 2 range from 0.1096 (C5) to 0.1149 (C8), a spread of less than 5%. All conditions show mild

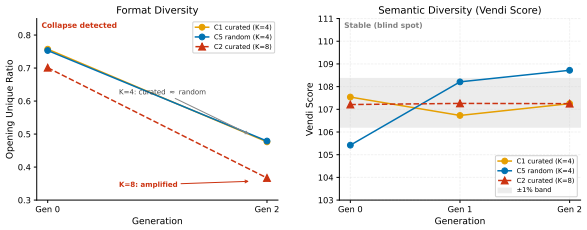


Figure 4: Format diversity metrics across generations for LoRA K=4 curated (C1) and random baseline (C5). Both conditions show nearly identical format degradation, confirming that format collapse is driven by iterative self-training rather than reward curation.

quality decreases from Gen 0 to Gen 2 (-0.3% to -1.3%; full per-generation scores in Table 7). The absence of meaningful quality improvement reflects saturation in the candidate quality distribution: the score gap between top-1 and average candidate is approximately 0.04, too narrow for the fine-tuned model to learn a meaningfully different quality level. The combined picture is that reward-based curation operates in a narrow effect regime at this scale: it does not collapse diversity, does not measurably improve quality, and its relative diversity effect depends more on training capacity than on curation strategy.

5 Analysis

5.1 Metric-Dependent Collapse: Format vs. Semantic

Diversity collapse under iterative self-training is real but mischaracterized by standard embedding-based metrics. Semantic diversity as measured by Vendi Score remains stable across generations (Section 4.1), yet surface-level formatting undergoes substantial homogenization. Table 4 presents format diversity metrics for the LoRA K=4 curated condition (C1) and the LoRA K=4 random baseline (C5) across Gen 0 and Gen 2. Opening unique ratio, defined as the fraction of responses with a unique first 50 characters, drops from 0.757 to 0.477 under curation and from 0.753 to 0.479 under the random baseline. Opening entropy (Shannon entropy over first-10-token sequences, in bits) declines by 0.75 bits under both conditions. The near-duplicate start rate (fraction of responses sharing a 50-character prefix with at least one other response) rises from 37.9% to 75.6% under curation and from 38.3% to 74.9% under the baseline. Both conditions exhibit nearly identical degradation trajectories (Figure 4), which indicates that

Table 4: Format diversity metrics for LoRA K=4 curated (C1) and LoRA K=4 random baseline (C5). Both conditions show nearly identical degradation across all metrics, which confirms that format collapse is driven by iterative self-training rather than reward curation.

Metric	C1 Gen 0	C1 Gen 2	C5 Gen 0	C5 Gen 2
Opening unique ratio	0.757	0.477	0.753	0.479
Opening entropy (bits)	12.43	11.68	12.43	11.68
Near-dup start rate (%)	37.9	75.6	38.3	74.9
Per-response TTR	0.682	0.670	0.683	0.660
Trigram repetition rate	0.174	0.256	0.174	0.258

format collapse is an intrinsic property of iterative self-training rather than a consequence of reward curation. The C2 condition (K=8) provides further evidence: its opening entropy drops by 1.13 bits, compared to 0.75 bits for both K=4 conditions. Doubling K amplifies homogenization, which suggests that candidate pool size acts as a scaling factor for format collapse. More candidates per prompt increase the probability of selecting responses that share common structural patterns and thereby reinforce template convergence across generations.

The mechanism is self-reinforcing: when a model trains on its own outputs, frequent structural patterns are overrepresented in the training data, the fine-tuned model generates these patterns more often, and the cycle repeats. This loop operates on surface formatting while leaving semantic content untouched, regardless of whether training data is reward-curated or random. Per-response type-token ratio (TTR) drops by only 0.012 under curation compared to 0.023 under random, confirming that the effect is confined to structural formatting rather than vocabulary.

The Vendi Score's insensitivity has a technical explanation: the sentence embeddings (all-MiniLM-L6-v2) encode semantic content and are largely invariant to surface formatting. Two responses with identical openings but different topics receive distant embeddings. This disconnect means Vendi Score alone provides an incomplete picture of how iterative training reshapes the output distribution.

5.2 Why Semantic Collapse Does Not Occur

We hypothesize three contributing factors that together account for the observed semantic stability. First, **prompt diversity** acts as an external anchor: 2,200 diverse prompts force broad semantic coverage at every generation regardless of per-prompt selection. Second, **reward model saturation** limits selection pressure. The ArmoRM score gap between top-1 and pool average is approximately 0.04, so the curated candidate differs only marginally from a random alternative. Third, **generation stochasticity** (T=0.7) injects fresh variation at each iteration, counteracting convergence to fixed outputs.

These factors connect to the sufficient conditions for model collapse identified by Shumailov et al. (2024): progressive distribution narrowing requires that synthetic data fully replaces external data and that sampling does not introduce compensating variance. In our setup, prompt diversity violates the first condition (an external scaffold persists across generations), and stochastic decoding violates the second. Reward saturation further weakens the selection channel through which curation could compound. C8 (K=16, 6.25% retention) provides the strongest test: under four times the selection pressure of K=4 configurations, Vendi Score changes by only -0.19% while training loss falls to 0.097. Even when curation retains fewer than 1 in 15 candidates, the semantic distribution remains stable. Isolating the individual contribution of each factor (e.g., through controlled ablation of prompt diversity or decoding temperature) remains an important direction for future work.

5.3 LoRA as Capacity Bottleneck

The similarity of diversity outcomes between K=4 and K=8 LoRA conditions despite doubled selection pressure suggests that parameter capacity, not curation intensity, is the binding constraint under LoRA. The rank-16 adapter modifies only 1.67% of parameters, limiting how far any training signal can reshape the output distribution (see Appendix B for detailed analysis).

5.4 Implications for Diversity Measurement

Standard diversity evaluation relies on embedding-based metrics that are insensitive to format-level homogenization. Our results show this workflow misses an entire dimension of collapse visible to human readers.

We propose two complementary metrics: *opening entropy* (Shannon entropy over first-10-token sequences) and *near-duplicate start rate* (fraction sharing a 50-character prefix). Both require only string operations, need no GPU, and capture a failure mode orthogonal to semantic metrics. In our experiments, opening entropy drops 0.75 bits at K=4 (1.13 bits at K=8) while Vendi Score changes by only -0.27%, and near-duplicate start rate rises 38

pp while Self-BLEU remains below 0.007.

The broader lesson is that "diversity" in iterative self-training is not a single quantity. Format, semantic, and lexical diversity can evolve independently under the same procedure. A complete diversity audit should include at least one metric from each dimension.

6 Discussion

6.1 Practical Implications

Existing diversity-preserving methods (DIVE (Qin et al., 2025), DARLING (Li et al., 2025a), GEM (Li et al., 2025b)) target semantic diversity as their primary objective, assuming that reward optimization narrows the semantic distribution. Our results challenge this assumption: across eight configurations, Vendi Score remains within $\pm 1\%$ of its initial value. The collapse that does occur is structural. Opening entropy drops by 0.75 bits under both curated and random conditions at $K{=}4$, intensifying to 1.13 bits at $K{=}8$. The near-duplicate start rate rises from 37.9% to 75.6% regardless of curation strategy. A practitioner who monitors only Vendi Score or Self-BLEU will conclude diversity is intact, while end users encounter structurally uniform responses.

Users perceive repetition through surface patterns, not embedding geometry. We recommend that diversity audits include at least one format-level metric alongside embedding-based measures. Opening entropy and near-duplicate start rate require no GPU and capture homogenization that current pipelines miss.

A separate practical constraint concerns the reward signal itself. The three proxy dimensions used in evaluation (quality, diversity, complexity) correlate at $r = 0.63$–0.77, limiting the effectiveness of single-objective curation. This entanglement explains both the near-zero quality improvement across generations and, paradoxically, why curation does not collapse semantic diversity: a reward signal correlated with multiple text properties selects for a broader band of responses than one isolating a single dimension. If the same reward model that curates training data also serves as the quality metric, improvements are circular; independent evaluation through held-out judges or orthogonal metrics remains essential.

6.2 Methodological Considerations

The theoretical conditions for model collapse (Shumailov et al., 2024) assume unfiltered self-consumption where all generated data is recycled. Reward-curated pipelines violate two preconditions: the training distribution is filtered by an external reward signal (not the model's full output), and the prompt set provides an external scaffold forcing broad semantic coverage. Reward saturation further limits compounding, with a score gap of only 0.04 between the top-1 candidate and pool average. These conditions are sufficient for semantic stability in our setting, but they are not guaranteed in all deployments. A narrow prompt domain would weaken the external scaffold, a highly discriminative reward model would sharpen selection pressure beyond the saturation regime we observe, and deterministic decoding would eliminate the variance injection that counteracts convergence. Practitioners operating in any of these regimes should verify semantic stability empirically rather than relying on our results.

The direction reversal between LoRA and Full-FT has broader methodological implications. Under LoRA, curation suppresses Vendi Score by 3.40 pp relative to random; under Full-FT at the same learning rate, the gap shrinks to 0.46 pp; at higher learning rate, curation promotes diversity by 0.63 pp. This progression is consistent with LoRA's rank-16 bottleneck imposing an upper bound on the distributional structure the model can absorb. Conclusions drawn from LoRA-based evaluations of curation may conflate capacity constraints with method properties. When the research question concerns distributional effects of data selection, evaluation under full fine-tuning is necessary to separate capacity effects from genuine curation dynamics.

7 Conclusion

This work systematically investigates reward curation effects on diversity in iterative self-training across 8 configurations and 3 generations. Our experiments reveal three findings. First, semantic diversity remains stable under all curated conditions (Vendi Score Δ: -0.27% to $+0.91\%$), extending to extreme selection pressure ($K{=}16$, $\Delta = -0.19\%$). Second, format diversity collapses sharply regardless of curation strategy (opening entropy -0.75 bits; near-duplicate rate 37.9% $\rightarrow$ 75.6%), yet standard embedding-based metrics are

blind to this degradation. Third, the diversity suppression observed under LoRA (-3.40 pp) reverses under Full-FT ($+0.63$ pp), a pattern consistent with LoRA's rank-16 bottleneck constraining distributional absorption rather than curation itself suppressing diversity.

For practitioners: semantic diversity safeguards appear unnecessary for standard iterative curation; the real risk is format homogenization requiring dedicated monitoring. Conclusions from LoRA-based experiments should not generalize to full-parameter training without verification. Scaling to larger models, longer generation chains, and format-aware selection methods remain open directions.

Limitations

Our findings are subject to several scope constraints that bound their generalizability.

Model Scale. All experiments use Qwen3-0.6B (596M parameters). Larger models produce higher-quality candidates with greater score variance, which could amplify curation effects in either direction: stronger selection pressure from wider score gaps, or greater resilience from richer internal representations. Whether the semantic stability we observe holds at the 7B or 70B scale remains an open question.

Generation Depth. We evaluate three generations of iterative self-training. The model collapse literature typically requires more iterations before distributional narrowing becomes severe (Shumailov et al., 2024). Our results establish that reward curation is safe over three generations, but they do not guarantee stability at generation ten or beyond. Subtle compounding effects below our measurement threshold could accumulate over longer horizons.

Domain Coverage. The UltraFeedback prompt set (2,200 instructions) covers general instruction-following across diverse topics. Narrow domains such as code generation or mathematical reasoning impose tighter output constraints and may exhibit different collapse dynamics. In particular, domains where correct answers cluster tightly in embedding space could reduce the protective effect of prompt diversity.

Single Reward Model. We use ArmoRM-Llama3-8B-v0.1 as the sole reward signal throughout all experiments. Different reward models encode different biases: some may penalize structural variety more aggressively, others may exhibit sharper score distributions that strengthen curation pressure. The specific format collapse patterns we document (opening template convergence, near-duplicate starts) reflect ArmoRM's preferences and may not generalize to all reward models.

Format Metric Granularity. Opening entropy and near-duplicate start rate capture homogenization in the first 10 tokens and 50 characters of each response, respectively. These metrics are blind to structural convergence in middle sections, concluding paragraphs, or organizational patterns such as list usage and paragraph structure. A complete characterization of format collapse would require finer-grained positional analysis that our current metrics do not provide.

References

Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoohi, and Richard G. Baraniuk. 2024. Self-Consuming Generative Models Go MAD. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*.

Ganqu Cui, Lifan Yuan, Ning Ding, Guanming Yao, Bingxiang He, Wei Zhu, Yuan Ni, Guotong Xie, Ruobing Xie, Yankai Lin, Zhiyuan Liu, and Maosong Sun. 2024. UltraFeedback: Boosting Language Models with Scaled AI Feedback. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.

Dan Friedman and Adji Bousso Dieng. 2023. The Vendi Score: A Diversity Evaluation Metric for Machine Learning. *Transactions on Machine Learning Research*.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *Proceedings of the Tenth International Conference on Learning Representations (ICLR)*.

Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. 2024. Understanding the Effects of RLHF on LLM Generalisation and Diversity. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*.

Tianjian Li, Yiming Zhang, Ping Yu, Swarnadeep Saha, Daniel Khashabi, Jason Weston, Jack Lanchantin, and Tianlu Wang. 2025a. Jointly Reinforcing Diversity and Quality in Language Model Generations. *arXiv preprint arXiv:2509.02534*.

Ziniu Li, Congliang Chen, Tian Xu, Zeyu Qin, Jiancong Xiao, Ruoyu Sun, and Zhi-Quan Luo. 2025b. Preserving Diversity in Supervised Fine-Tuning of Large Language Models. In *Proceedings of the Thirteenth International Conference on Learning Representations (ICLR)*.

Ilya Loshchilov and Frank Hutter. 2019. Decoupled Weight Decay Regularization. In *Proceedings of the Seventh International Conference on Learning Representations (ICLR)*.

Yiwei Qin, Yixiu Liu, and Pengfei Liu. 2025. DIVE: Diversified Iterative Self-Improvement. *arXiv preprint arXiv:2501.00747*.

Qwen Team. 2025. Qwen3 Technical Report. *arXiv preprint arXiv:2505.09388*.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. 2024. AI models collapse when trained on recursively generated data. *Nature*, 631:755–759.

Haoran Sun, Yekun Chai, Shuohuan Wang, Yu Sun, Hua Wu, and Haifeng Wang. 2025. Curiosity-Driven Reinforcement Learning from Human Feedback. *arXiv preprint arXiv:2501.11463*.

Haoxiang Wang, Wei Xiong, Tengyang Xie, Han Zhao, and Tong Zhang. 2024. Interpretable Preferences via Multi-Objective Reward Modeling and Mixture-of-Experts. *arXiv preprint arXiv:2406.12845*.

Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. 2020. MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*.

Yaoming Zhu, Sidi Lu, Lei Zheng, Jiaxian Guo, Weinan Zhang, Jun Wang, and Yong Yu. 2018. Texygen: A Benchmarking Platform for Text Generation Models. In *Proceedings of the 41st International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1097–1100.

A Training Details

Table 5 lists the complete hyperparameter settings for all experimental conditions.

Table 5: Hyperparameters for all experimental conditions.

Parameter	LoRA	Full-FT
Base model	Qwen3-0.6B	
Reward model	ArmoRM-Llama3-8B-v0.1	
Prompts (train / eval)	2000 / 200	
Generations	3	
Temperature	0.7	
Max new tokens	1024	
Epochs per generation	2	2 (C4, C8: 3)
Per-device batch size	4	1
Gradient accumulation	4	8
Optimizer	AdamW	
Weight decay	0.01	
Warmup ratio	0.1	
Precision	bf16	
Learning rate	2×10^{-4}	5×10^{-6} / 2×10^{-5}
LoRA rank	16	–
LoRA alpha	32	–
LoRA target	q/k/v/o_proj, gate/up/down_proj	–
Gradient checkpointing	No	Yes

B LoRA as Capacity Bottleneck

The similarity of diversity outcomes between $K{=}4$ and $K{=}8$ LoRA conditions (C1: -0.27%, C2: $+0.04\%$) despite doubled selection pressure suggests that parameter capacity, not curation intensity, is the binding constraint under LoRA. If curation were the primary driver of diversity change, doubling the candidate pool from 4 to 8 (reducing the retention rate from 25% to 12.5%) should produce measurably greater diversity suppression. Instead, both conditions converge to nearly identical Gen 2 Vendi Scores (107.25 for both C1 and C2). This convergence is consistent with a capacity-limited regime in which the adapter cannot express the differences between curated and non-curated training signals.

The rank-16 LoRA adapter modifies only 1.67% of model parameters. All learned changes, whether driven by reward-curated data or random data, must be expressed within this low-dimensional subspace. Training loss provides direct evidence of this constraint: C1 (curated, $K{=}4$) reaches a final loss of 0.166, C2 (curated, $K{=}8$) reaches 0.168, and C5 (random) reaches 0.152. The 0.014 gap between curated and random final loss is small relative to the 0.240 reduction from Gen 0 to Gen 2 within each condition (Table 3). The adapter absorbs most of the training signal's structure regardless of whether that signal comes from curated or random selection; the residual difference in loss translates to a negligible difference in diversity outcomes.

This interpretation reframes the 3.40 pp curation gap reported in Section 4.3. The gap does not reflect curation actively suppressing diversity. Rather, random data's uncorrelated gradients happen to push the LoRA subspace in directions that slightly increase output diversity, while curated data's correlated gradients (consistent reward-preferred patterns) do not. Under full fine-tuning, where the capacity constraint is removed, this asymmetry disappears and eventually reverses (Section 4.3). The practical implication is that LoRA-based studies of reward curation effects should be interpreted cautiously. Observed diversity changes under

parameter-efficient fine-tuning may reflect adapter capacity limits rather than properties of the curation strategy itself.

C Per-Generation Metrics

Table 6 reports four diversity and fluency metrics for every condition at each self-training generation. Vendi Score measures semantic diversity via matrix-rank entropy over embedding similarities; higher values indicate a more diverse output distribution. Self-BLEU averages pairwise BLEU between generated responses; lower values indicate less repetition. Perplexity (PPL) is computed with a held-out language model; stable values indicate no fluency degradation. Nearest-neighbor distance (NND) is the mean cosine distance to the closest other response in embedding space; higher values reflect greater spread.

D ArmoRM Quality Scores

Table 7 reports mean ArmoRM quality scores per generation for all conditions, computed on 200 held-out evaluation prompts.

E Lexical Diversity: Self-BLEU and Distinct-n

Table 8 reports Self-BLEU-4 (computed over 500 randomly sampled responses per generation, seed 42) and Distinct-n ($n \in \{1, 2, 3\}$, computed over all 2,000 training responses). Distinct-n is defined as the ratio of unique n-grams to total n-grams across the response corpus.

Table 6: Per-generation metrics for all 8 experimental conditions across 3 self-training generations. Curated conditions (C1–C4, C8) show stable Vendi Score ($\pm 1\%$), low Self-BLEU, and consistent NND, indicating no semantic collapse. Random baselines (C5–C7) exhibit comparable or larger metric drift despite the absence of reward selection. C7 metrics were recorded from a separate analysis pass.

Condition	Gen	Vendi Score	Self-BLEU (↓)	PPL	NND
Curated conditions					
C1: LoRA, $K{=}4$	0	107.54	0.0059	10.60	0.7088
	1	106.73	0.0091	9.73	0.7065
	2	107.25	0.0062	9.72	0.7054
C2: LoRA, $K{=}8$	0	107.21	0.0057	9.40	0.7097
	1	107.26	0.0024	9.18	0.7086
	2	107.25	0.0064	10.20	0.7095
C3: Full-FT, 5e-6	0	107.10	0.0038	9.10	0.7073
	1	107.61	0.0060	8.66	0.7091
	2	108.07	0.0044	9.21	0.7108
C4: Full-FT, 2e-5	0	107.81	0.0046	9.57	0.7079
	1	108.03	0.0039	10.34	0.7107
	2	108.54	0.0040	8.82	0.7100
C8: Full-FT, $K{=}16$	0	107.32	0.0057	8.50	0.7057
	1	106.73	0.0050	8.02	0.7065
	2	107.11	0.0058	7.71	0.7092
Random baselines					
C5: LoRA, $K{=}4$	0	105.42	0.0044	9.19	0.7005
	1	108.21	0.0067	8.76	0.7121
	2	108.72	0.0067	11.18	0.7136
C6: Full-FT, 5e-6	0	107.17	0.0045	8.77	0.7061
	1	107.87	0.0047	8.70	0.7094
	2	108.64	0.0065	8.82	0.7114
C7: Full-FT, 2e-5	0	108.08	0.0046	10.97	0.7064
	1	107.85	0.0049	10.83	0.7095
	2	108.13	0.0068	11.15	0.7094

Table 7: Mean ArmoRM quality score per generation. All conditions remain in the 0.110–0.116 range with mild decreases across generations (-0.3% to -1.3%), confirming quality stability under iterative self-training.

	Config	Gen 0	Gen 1	Gen 2	Δ
Curated					
C1	LoRA, $K{=}4$	0.1133	0.1132	0.1142	$+0.08\%$
C2	LoRA, $K{=}8$	0.1146	0.1136	0.1143	-0.03%
C3	Full-FT, 5e-6	0.1145	0.1141	0.1131	-1.2%
C4	Full-FT, 2e-5	0.1130	0.1154	0.1124	-0.5%
C8	Full-FT, $K{=}16$	0.1162	0.1162	0.1149	-1.1%
Random baseline					
C5	LoRA, $K{=}4$	0.1107	0.1099	0.1096	-1.0%
C6	Full-FT, 5e-6	0.1126	0.1111	0.1117	-0.8%
C7	Full-FT, 2e-5	0.1112	0.1100	0.1097	-1.3%

Table 8: Self-BLEU-4 and Distinct-n across generations. Self-BLEU-4 remains low ($<$0.016) for all conditions, indicating minimal textual repetition. Distinct-n shows uniform decline across all conditions (curated and random alike), consistent with format homogenization under iterative self-training rather than reward-specific narrowing.

Condition	Gen	Self-BLEU-4 (↓)	Distinct-1	Distinct-2	Distinct-3
Curated conditions					
C1: LoRA, K=4	0	0.0116	0.167	0.582	0.835
	1	0.0160	0.163	0.560	0.808
	2	0.0137	0.157	0.546	0.793
C2: LoRA, K=8	0	0.0138	0.167	0.579	0.831
	1	0.0144	0.164	0.562	0.810
	2	0.0149	0.162	0.555	0.802
C3: Full-FT, 5e-6	0	0.0106	0.166	0.579	0.832
	1	0.0144	0.165	0.569	0.819
	2	0.0158	0.162	0.556	0.803
C4: Full-FT, 2e-5	0	0.0131	0.168	0.580	0.833
	1	0.0117	0.162	0.562	0.811
	2	0.0101	0.160	0.557	0.807
C8: Full-FT, K=16	0	0.0120	0.170	0.581	0.832
	1	0.0156	0.165	0.570	0.819
	2	0.0160	0.163	0.562	0.811
Random baselines					
C5: LoRA, K=4	0	0.0107	0.168	0.580	0.831
	1	0.0136	0.158	0.545	0.789
	2	0.0132	0.156	0.534	0.776
C6: Full-FT, 5e-6	0	0.0133	0.169	0.582	0.833
	1	0.0133	0.165	0.563	0.811
	2	0.0121	0.162	0.550	0.795
C7: Full-FT, 2e-5	0	0.0133	0.167	0.580	0.832
	1	0.0138	0.162	0.556	0.800
	2	0.0141	0.157	0.541	0.783